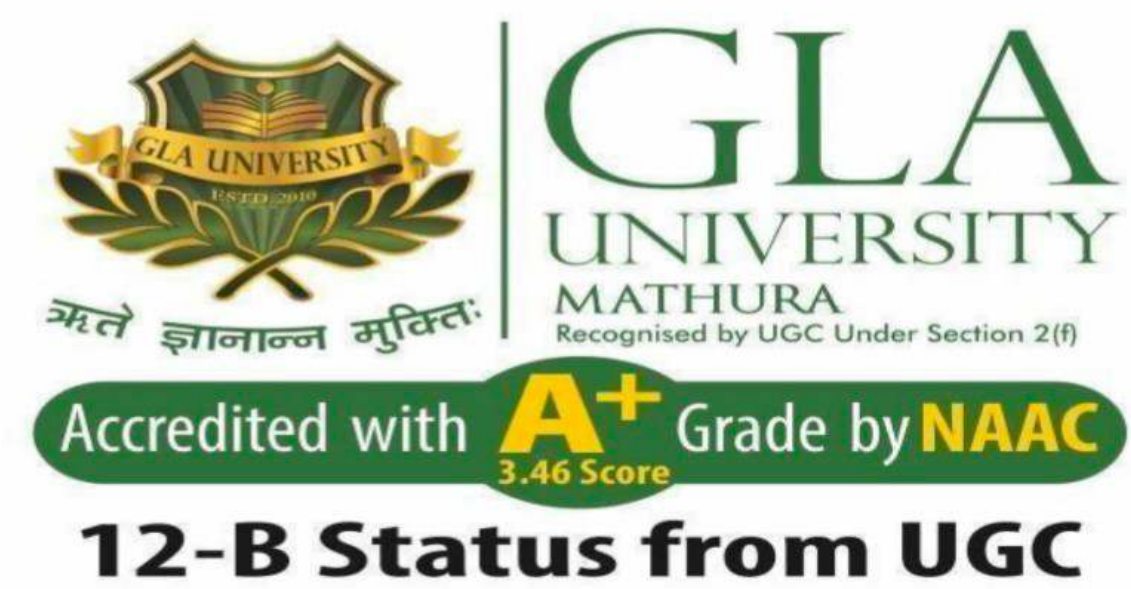
B.Tech CSE (III YEAR – VI SEM) (2025-26)

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING & APPLICATIONS



GLA UNIVERSITY

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Project Title: Smart Body Posture Recognition & Guidance System

Team Leader:	Prasant Kumar	UR- 2415500346
Team Member 1:	Pawan Kumar	UR- 2415500334
Team Member 2:	Dhananjay Chaudhary	UR - 2415500155
Mentor Name:	**Rohit kumar gupta sir**	

Signature :

Project Synopsis

Project title: Smart Body Posture Recognition & Guidance System
Team name & ID: OG-Coders (T-89)
Institute / Course: GLA University, Mini-Project
Version: v1.0
Date: 02 Sep 2026

1. Overview

Problem statement: In self-service health kiosks and diagnostic stations, accurate measurement of vital signs (blood pressure, weight, body composition, ECG) depends directly on proper body posture. Slouching, crossed legs, or poor arm alignment can distort readings — for example, sitting slouched can falsely elevate blood pressure readings by 10–15 mmHg or degrade ECG signal quality. Unattended users frequently position themselves incorrectly, forcing staff to intervene manually, while kiosks that trigger measurements regardless of posture produce failed scans, invalid records, and user frustration.

Goal: Build an AI-powered computer vision module that connects to a camera feed, detects human body keypoints in real time, compares them against correct posture rules, provides instant audio/visual correction guidance, and automatically starts health measurements only once the user is properly aligned.

Non-goals: Direct control or certification of medical sensor hardware, clinical diagnosis of health conditions, and multi-camera 3D body reconstruction in v1.

Value proposition: A real-time, self-correcting posture guidance layer that sits in front of any vitals-measuring kiosk — removing staff dependency, cutting positioning errors from ~30% to under 2%, and reducing average check-in time to under 2 minutes.

2. Scope and Control

2.1 In-scope

- Real-time human pose/keypoint detection from a live camera feed
- Rule-based posture validation for seated and standing vitals checks
- On-screen visual overlay showing posture errors
- Voice/audio prompts guiding the user to correct posture
- Automatic sensor-trigger signal once correct posture is held for 2–3 seconds
- Anonymized posture-event logging for analytics

2.2 Out-of-scope

- Actual blood pressure/ECG/weight sensor hardware integration logic
- Facial recognition or user identity tracking
- Mobile app version

2.3 Assumptions

- Kiosk camera has a clear, adequately lit view of the user
- One user is present at the kiosk at a time
- Users can see/hear on-screen and audio prompts

2.4 Constraints

- Limited access to real clinical posture/vitals datasets
- Camera hardware and lighting variability across kiosk sites
- Academic project timeline

2.5 Dependencies

- Pose-estimation model/library (e.g., MediaPipe Pose / OpenPose)
- Camera feed (USB webcam/kiosk-integrated camera)
- Text-to-speech engine for audio prompts

2.6 Acceptance criteria and sign-off

- GIVEN a user is seated at the kiosk WHEN posture deviates from the defined rules THEN a corrective visual and audio prompt is shown within 1 second
- GIVEN correct posture is detected WHEN it is held continuously for 2-3 seconds THEN the sensor-activation signal is triggered automatically
- System must sustain real-time frame processing at an acceptable frame rate (≥15 FPS) on kiosk-grade hardware

Stakeholder	Role	Decision area	Signature/Approval	Date
Rohit Kumar Gupta Sir	Mentor	Scope, final acceptance	Approved	02 Sep 2026
Prasant Kumar	Product Lead	Release readiness	Approved	02 Sep 2026

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Pawan Kumar	Prasant Kumar	Mentor	Team
Design	Pawan Kumar & Dhananjay Chaudhary	Pawan Kumar & Dhananjay Chaudhary	Mentor	Team
Implementation	Prasant Kumar	Prasant Kumar		Team
Testing	Dhananjay Chaudhary	Prasant Kumar	Mentor	Team
Release	Dhananjay Chaudhary	Dhananjay Chaudhary	Mentor	Dept

4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Pawan Kumar	Product Lead	Scope, backlog, reviews	Product, CV pipelines	8 hrs/wk	pawan.kumar_cs.aiml24@gla.ac.in
Prasant Kumar & Pawan Kumar	Tech Lead & CV/Backend	Pose-model integration, rules engine, APIs	Python, OpenCV, MediaPipe, Flask	10 hrs/wk	prasant.kumar_cs.aiml24@gla.ac.in
Prasant Kumar & Dhananjay Chaudhary	UI/Kiosk Frontend	On-screen overlay, audio prompts, kiosk UI	React/PyQt, TTS	10 hrs/wk	prasant.kumar_cs.aiml24@gla.ac.in
Dhananjay Chaudhary	QA & Docs	Test plan, accuracy validation, docs	Testing, writing	8 hrs/wk	dhananjay.chaudhary_cs.aiml24@gla.ac.in

5. Week-wise Plan and Assignments

Week	Dates	Milestones	Prasant (Lead)	Prasant (CV/Backend)	Dhananjay (UI)	Pawan (QA/Docs)	Deliverables
1	2-8 Sep	Requirements freeze	Finalize scope	Pose-model selection	Wireframes	Test plan v0	Draft SRS
2	9-15 Sep	Architecture & pipeline	Arch review	Keypoint pipeline setup	UI kit	Test data	Architecture v1
3	16-22 Sep	Posture rule engine	Risks, unblock	Rule engine (seated posture)	Overlay screens	Smoke tests	Rule engine module
4	23-29 Sep	Audio/visual feedback	Sync	TTS integration	Overlay + audio UI	Accessibility checks	Feedback module
5	30 Sep-6 Oct	Auto-trigger logic	Cut scope	Hold-timer & trigger signal	Kiosk screen flow	Functional tests	Feature demo A
6	7-13 Oct	Accuracy tuning	KPIs	Rule calibration, edge cases	UI polish	Accuracy tests	Feature demo B
7	14-20 Oct	Hardening	Risk burndown	Bug fixes	Bug fixes	Regression suite	Test report
8	21-27 Oct	Release & deck	Final sign-off	Release notes	Polish UI	User manual	v1.0, slides

6. Users and UX

6.1 Personas

- Elderly/First-time Kiosk User: Needs simple, patient, step-by-step posture guidance
- Pharmacy/Clinic Walk-in Patient: Wants a fast, frustration-free check-in without staff help

6.2 Top User Journeys

User approaches kiosk → Camera detects presence → System checks posture in real time → Visual/audio guidance corrects positioning → Correct posture held for 2-3 seconds → Sensor scan triggers automatically → Reading recorded

6.3 User Stories

As a kiosk user, I want clear real-time posture guidance so that my vitals are measured accurately without needing staff assistance.

6.4 Accessibility & Localization

- Simple, jargon-free voice prompts
- Large, high-contrast on-screen overlay indicators
- English language support (v1); regional language support planned for future versions

7. Market and Competitors

7.1 Competitor table

Competitor	Product	Target users	Key features	Pricing	Strengths	Weaknesses
Withings Body Cardio Kiosks	Smart health kiosks	Pharmacies, clinics	Weight, BP, basic vitals capture	Enterprise licensing	Established hardware, brand trust	No active posture correction; relies on user self-positioning
Higi Smart Health Stations	Self-service health kiosks	Retail pharmacies	BP, weight, BMI tracking	Per-kiosk licensing	Wide retail deployment, simple UX	Static instructions only; no live feedback loop
OpenPose/MediaPipe-based research	Open-source pose libraries	Researchers, developers	Generic keypoint detection	Free/open-source	Flexible, well-documented	No domain-specific posture rules or kiosk integration

7.2 Positioning

The system focuses on being a real-time, explainable posture-correction layer that plugs into existing health kiosks turning unreliable self-positioning into automated, clinical-grade alignment.

Measurable Delta:

- Positioning error rate: <2% vs ~30% (traditional self-service kiosks)
- Check-in time: <2 min vs 4+ min (manual/staff-assisted)
- Staff dependency: 0 vs continuous supervision (current kiosks)

8. Objectives and Success Metrics

- O1: Posture detection latency - Real-time keypoint detection at ≥15 FPS by 30 Sep 2026 (KPI: frames/sec)

- O2: Correction accuracy - ≥95% correct classification of posture errors by 15 Oct 2026 (KPI: classification accuracy)
- O3: Auto-trigger reliability - ≥98% of correctly-postured sessions trigger the scan signal by 20 Oct 2026 (KPI: trigger success rate)
- O4: Error reduction - Positioning-error-driven invalid readings reduced to <2% by release (KPI: error rate)

9. Key Features

Feature	Description	Priority	Dependencies	Acceptance criteria
Real-Time Pose Detection	Detects human body keypoints (shoulders, spine, arms, legs) from the live camera feed	Must	Camera feed	GIVEN a user is in frame WHEN the camera streams video THEN keypoints are detected within 200ms
Posture Rule Engine	Compares detected keypoints against defined correct-posture rules for the specific vitals check	Must	Pose Detection	GIVEN keypoints are extracted WHEN compared to posture rules THEN a pass/fail posture status is returned
Visual Overlay Feedback	Displays on-screen indicators highlighting which body part needs adjustment	Must	Rule Engine	GIVEN a posture error is detected WHEN feedback renders THEN the affected body part is visually highlighted within 1 second
Voice Guidance	Provides spoken instructions to correct posture (e.g., 'Please sit up straight')	Must	Rule Engine	GIVEN a posture error persists WHEN 2 seconds elapse THEN an audio prompt is triggered
Auto Sensor-Trigger	Automatically signals the vitals sensor to start once correct posture is held steadily	Must	Rule Engine, Timer	GIVEN correct posture is detected WHEN held for 2–3 seconds THEN the sensor-activation signal fires

Posture Analytics Logging	Stores anonymized posture event data to identify common user mistakes	Should	Storage	GIVEN a session ends WHEN posture events occurred THEN they are logged anonymously for analysis

10. Architecture

10.1 High-Level Architecture

The system follows a modular, real-time processing architecture designed to run on kiosk-grade edge hardware with minimal latency.

- Client / Kiosk Interface: Touchscreen kiosk display showing live camera preview, posture overlay, and status indicators, paired with a speaker for audio prompts.
- Core Processing Services:
- Video Capture Service: Streams frames from the kiosk camera in real time
- Pose Estimation Engine: Runs a keypoint-detection model (e.g., MediaPipe Pose) on each frame
- Posture Rule Engine: Evaluates keypoint geometry (joint angles, alignment) against posture rule sets per measurement type
- Feedback Controller: Drives the visual overlay and text-to-speech audio guidance
- Trigger Service: Monitors the hold-timer and emits the sensor-activation signal to the connected vitals device
- Data Stores:
- Local/Cloud Database: Stores anonymized posture session logs and analytics
- Config Store: Holds posture rule definitions per measurement type (BP, weight, ECG)
- Integrations:
- Kiosk vitals sensors (BP cuff, weighing scale, ECG unit) via a trigger/API signal
- Text-to-speech engine for audio prompts

10.2 API spec snapshot

Endpoint	Method	Auth	Purpose	Request schema	Response schema
/api/posture/analyze	POST	—	Analyze current frame keypoints against rules	frame/keypoints, measurementType	200 {status, errors[]}
/api/posture/trigger	POST	Device key	Signal sensor to start measurement	sessionId	200 {triggered:true}
/api/analytics/log	POST	—	Log anonymized posture session event	sessionId, errorTypes[]	201 {logId}

10.3 Configuration and Secrets

- Environment variables managed using .env files
- Posture rule thresholds stored in versioned config files
- Device/API keys Git-ignored and rotated periodically
- Access restricted to authorized CI/CD pipelines

11. Data Design

11.1 Core Entities

- Session
- PostureEvent
- PostureRule
- MeasurementType
- AnalyticsLog

Posture and session data is anonymized at capture; no facial or identity data is stored.

11.2 Data dictionary

Entity	Field	Type	Null?	Allowed values	Source	Notes
Session	id	UUID	No	—	System	PK
PostureEvent	errorType	Enum	No	Slouch/CrossedLegs/ArmMisaligned	System	—
PostureRule	measurementType	Enum	No	BP/Weight/ECG	Config	PK
PostureEvent	timestamp	DateTime	No	—	System	Indexed
AnalyticsLog	sessionDurationMs	Integer	No	≥ 0	System	—

11.3 Schemas and Migrations

- ER Diagram included in appendix
- Database migrations version-controlled
- Rollback tested on staging environment

11.4 Privacy, Retention, Backup, and DR

- No Personally Identifiable Information (PII) is captured — only anonymized posture-event data
- Retention policy: raw video frames discarded immediately after processing; logs retained for 6 months
- Nightly automated backups of analytics logs
- RTO: 4 hours
- RPO: 24 hours

12. Technical Workflow Diagrams

The following diagrams are included to explain system behavior:

- State Transition Diagram (posture states: Not Detected → Incorrect → Correcting → Correct → Triggered)
- Sequence Diagram (camera → pose engine → rule engine → feedback → trigger)
- Use Case Diagram
- Data Flow Diagram (DFD)
- Entity Relationship Diagram (ERD)
- System Architecture Diagram

13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	SLI	Target (SLO)	Measurement
Availability	Uptime %	≥ 99.0%	Uptime monitor
Latency	Frame-to-feedback p95	≤ 300 ms	APM
Accuracy	Posture classification accuracy	≥ 95%	Validation dataset
Error rate	Failed auto-trigger %	≤ 2%	Logs
Security	Critical CVEs	0	Vulnerability scanner

13.2 Test plan

Area	Type	Tools	Owner	Coverage target	Exit criteria
Pose/Rule Engine	Unit	Pytest	Prasant	70%	No P1/P2 defects
Kiosk UI	E2E	Playwright	Dhananjay	60%	Pass rate ≥ 95%
API Integration	Postman	Pawan	All scenarios	All critical pass	

13.3 Environments

- Development → Staging → Kiosk Pilot
- Feature flags used for experimental posture rules

14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
Camera feed	Unauthorized access/interception	Information Disclosure	High	Low	Encrypted local processing, no cloud video storage	Prasant
Trigger API	Spoofed trigger signal	Spoofing	Medium	Low	Device-key auth, signed requests	Prasant

14.2 AuthN/AuthZ

- Device-key based authentication for sensor-trigger API
- Role-based access control for kiosk admin/config panel

14.3 Audit and logging

Log posture sessions, trigger events, and rule-config changes; 90-day retention

14.4 Compliance

Academic project; no PII captured; follows institute data-handling policy

15. Delivery and Operations

15.1 Release plan

- Version v1.0 demo at project submission
- Incremental posture-rule rollout per measurement type (BP → Weight → ECG)

15.2 CI/CD and rollback

- CI pipeline: lint → test → build → deploy
- Rollback using previous container/model image

15.3 Monitoring and alerting

Metric	Threshold	Alert to	Runbook
Frame-to-feedback latency	> 500 ms	Tech Lead	"Latency Spike" runbook
Auto-trigger failure rate	> 5%	Tech Lead	"Trigger Failure" runbook

15.4 Runbooks

- Latency Spike: check camera frame rate → reduce model resolution → restart pose service
- Trigger Failure: inspect posture-rule logs → verify sensor connection → roll back rule config

15.5 Communication plan

Standups Mon/Wed/Fri. Weekly status to mentor each Friday. Bi-weekly demo.

16. Risks and Mitigations

Risk	Probability	Impact	Score	Mitigation	Owner
Schedule slip	Medium	High	12	Scope freeze, weekly demos	Prasant
Poor lighting affecting detection accuracy	Medium	Medium	9	Adaptive brightness handling, IR camera fallback	Prasant
Posture rule false positives	Low	Medium	6	Threshold tuning on validation dataset	Prasant

17. Research and Evaluation

17.1 Study of Existing Platforms

Existing self-service health kiosks (e.g., Higi Smart Health Stations, Withings-based systems) focus on capturing vitals but rely on printed or static instructions for user positioning, with no real-time feedback loop. Open-source pose libraries like MediaPipe and OpenPose provide generic keypoint detection but lack domain-specific posture rules for clinical measurement contexts. This gap motivated the Smart Body Posture Recognition & Guidance System — a purpose-built layer that actively corrects posture before a scan begins.

17.2 Evaluation Using Test Data

The posture rule engine will be evaluated against a labeled dataset of seated postures (correct vs. common error types — slouching, crossed legs, arm misalignment) to measure classification accuracy and false-positive/false-negative rates across varied body types and camera angles.

17.3 User Feedback

Simulated kiosk users will test the visual overlay and audio guidance for clarity, responsiveness, and ease of correction, with feedback used to refine prompt wording and overlay design.

17.4 KPI Tracking

Key metrics include frame-to-feedback latency, posture classification accuracy, auto-trigger success rate, and overall reduction in invalid-reading rate — tracked continuously to guide iteration.

18. Appendices

18.1 Glossary

- AI: Systems simulating human intelligence
- Keypoint: A detected anatomical landmark (e.g., shoulder, elbow, hip) used for pose estimation
- KPI: Metrics measuring performance
- SLO: Target service performance
- p95: 95th percentile response time

18.2 References

- MediaPipe Pose: https://developers.google.com/mediapipe/solutions/vision/pose_landmarker
- OpenPose: https://github.com/CMU-Perceptual-Computing-Lab/openpose
- OpenCV: https://opencv.org
- React: https://react.dev
- MDN Web Docs: https://developer.mozilla.org
- Scikit-learn: https://scikit-learn.org
- IEEE: "Pose Estimation for Healthcare Applications" – https://ieeexplore.ieee.org
- Springer: "Computer Vision in Clinical Settings" – https://link.springer.com